P.7 MATHEMATICS

LESSON PLANS

TERM 1 2026

LESSON PLAN 1

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Subsets and Proper Subsets

COMPETENCES:

- **Subject Competence:** The learner defines subsets and proper subsets and lists all subsets of a given set.
- **Language Competence:** The learner uses terms like subset, proper subset, universal set correctly.

TEACHING METHODS/TECHNIQUES:

- Question and answer
- Demonstration
- Guided discovery
- Use of charts
- Discussion
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Define a subset and a proper subset
2. List all subsets of a given set
3. List all proper subsets of a given set

4. Identify the number of subsets and proper subsets

TEACHING/LEARNING MATERIALS:

- Chart showing subsets
- Flash cards with set members
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners warmly - Writes set P = {a, g, t} on the board - Asks: "Can we form smaller sets from this set?" - Asks learners to suggest possible smaller sets - Introduces topic: subsets and proper subsets	- Question and answer - Brainstorming - Demonstration	- Responds to greeting - Observes the set - Suggests smaller sets - Listens attentively
15 mins	**EXPLORE**	- Divides learners into groups - Gives each group set P = {a, g, t} - Asks them to list all possible smaller sets that can be formed - Asks: "How many sets have you formed? Which one is the original set?" - Lets groups share their findings	- Group work - Guided discovery - Discussion	- Works in groups - Lists smaller sets - Counts the sets formed - Shares findings with class
20 mins	**EXPLAIN**	- Defines **subset**: a set formed from the main/given set; a part of the universal set; a set contained in another set - Explains two types of subsets: - **Proper subset**: has fewer members than the universal set - **Improper subset**: a subset which is equal to the universal set - Lists all subsets of P = {a, g, t}: { }, {a}, {g}, {t}, {a,g}, {a,t}, {g,t}, {a,g,t}	- Lecture method - Demonstration - Use of examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions

		- Explains: every given set is a subset of itself; empty set is a subset of every set - Lists proper subsets: all subsets except the original set { }, {a}, {g}, {t}, {a,g}, {a,t}, {g,t} - Notes: original set is not regarded as a proper subset because proper subsets have fewer elements		
10 mins	**ELABORATE**	- Gives set T = {3, k} and asks learners to list all subsets and proper subsets - Guides them to find: subsets = { }, {3}, {k}, {3,k} (4 subsets); proper subsets = { }, {3}, {k} (3 proper subsets) - Asks learners to give more examples of sets and list their subsets - Discusses: "Why is the empty set considered a subset of every set?"	- Guided practice - Application - Question and answer	- Lists subsets and proper subsets - Answers questions - Participates in discussion
5 mins	**EVALUATE**	- Asks oral questions: - "What is a subset?" - "What is a proper subset?" - "How many subsets does a set with 3 elements have?" - "How many proper subsets does a set with 3 elements have?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to list all subsets of set H = {b, e, d} at home - Asks them to find out the formula for finding number of subsets - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. What is a subset?
2. List all the subsets of set B = {1, 3, 4}
3. List all the proper subsets of set F = {d, 4}
4. How many subsets does a set with 3 elements have?
5. How many proper subsets does a set with 2 elements have?

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 2

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Finding number of subsets and proper subsets using formula

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COMPETENCES:

- **Subject Competence:** The learner uses the formula 2^n to find the number of subsets and $2^n - 1$ to find the number of proper subsets.
- **Language Competence:** The learner explains the meaning of n in the formula and uses it correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Problem-solving
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. State the formula for finding number of subsets
2. State the formula for finding number of proper subsets
3. Calculate number of subsets using 2^n
4. Calculate number of proper subsets using $2^n - 1$
5. Find the number of elements when given number of subsets or proper subsets

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TEACHING/LEARNING MATERIALS:

- Chart with formulas
- Flash cards
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Reviews previous lesson: "How many subsets did set P = {a,g,t} have?" (8) - Asks: "Is there a faster way to find this without listing?" - Introduces topic: formulas for subsets and proper subsets	- Review - Question and answer - Brainstorming	- Responds to greeting - Answers review questions - Suggests possible formulas - Listens
15 mins	**EXPLORE**	- Gives groups sets with different numbers of elements: 1 element, 2 elements, 3 elements - Asks them to list all subsets and count them - Asks them to look for a pattern: 1 element → 2 subsets; 2 elements → 4 subsets;	- Group work - Discovery method - Pattern recognition	- Lists subsets for given sets - Counts subsets - Observes pattern: 2,

		3 elements → 8 subsets - Lets groups share their observations		4, 8 - Shares findings
20 mins	**EXPLAIN**	- Explains **formula for number of subsets**: 2^n where n = number of elements - Example: P = {a,g,t,y} has 4 elements → $2^4 = 16$ subsets - Example: Set with 5 elements → $2^5 = 32$ subsets - Explains **formula for number of proper subsets**: $2^n - 1$ - Proper subsets = all subsets minus the original set - Example: P = {a,g,t,y} → $2^4 - 1 = 16 - 1 = 15$ proper subsets - Example: Set with 5 elements → $2^5 - 1 = 32 - 1 = 31$ proper subsets - Shows how to find number of elements when given number of subsets: - If subsets = 16, then $2^n = 16 \rightarrow 2^n = 2^4 \rightarrow n = 4$ - If proper subsets = 31, then $2^n - 1 = 31 \rightarrow 2^n = 32 \rightarrow 2^n = 2^5 \rightarrow n = 5$	- Lecture method - Formula explanation - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies formulas - Asks questions
10 mins	**ELABORATE**	- Provides practice problems: a) Find number of subsets in set B = {1,3,4,6} b) Find number of proper subsets in set H = {b,e,a,d,s} c) How many elements are in a set with 8 subsets? d) How many elements are in a set with 63 proper subsets? - Guides learners through solutions	- Guided practice - Problem-solving - Application	- Solves problems - Shows working - Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "What is the formula for number of subsets?" - "What is the formula for number of proper subsets?" - "How many subsets does a set with 6 elements have?" - "How many proper subsets does a set with 6 elements have?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to find the number of elements in a set with 128 subsets - Asks them to practice more problems at home - Prepares for next lesson on Venn diagrams	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. How many subsets can be got from a set with 6 elements?
2. How many proper subsets can be got from a set with 6 elements?
3. Find the number of subsets in set K = {a,b,c,d,e,f,g,h,i,j}
4. Find the number of proper subsets in set L = {a,e,i,o,u}
5. How many elements are in a set with 32 subsets?
6. How many elements are in a set with 31 proper subsets?

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 3

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Seven	Mathematics		60 Minutes

TOPIC	**SUB-TOPIC**	**CONTENT**
Sets	Number Patterns	Parts of a Venn Diagram and Describing Shaded Regions

COMPETENCES:

- **Subject Competence:** The learner identifies and describes different regions in a Venn diagram.
- **Language Competence:** The learner uses terms like intersection, union, complement, only correctly.

TEACHING METHODS/TECHNIQUES:

- Use of charts
- Demonstration
- Question and answer
- Drawing
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Identify different parts of a Venn diagram
2. Describe shaded regions using set notation
3. Shade given regions on a Venn diagram

TEACHING/LEARNING MATERIALS:

- Chart showing Venn diagrams with different shaded regions
- Blank Venn diagrams for practice
- Colored chalk
- Chalkboard/whiteboard

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Draws two intersecting circles on the board - Asks: "What do you call this diagram?" (Venn diagram) - Asks: "How many regions can you see?" (4 regions) - Introduces topic: parts of a Venn diagram	- Question and answer - Drawing - Brainstorming	- Responds to greeting - Observes diagram - Identifies regions - Listens
15 mins	**EXPLORE**	- Draws a Venn diagram with two sets P and Q - Shades different regions and asks groups to describe each in their own words - Regions: P only, Q only, intersection, union, complement of P, etc. - Lets groups share their descriptions	- Group work - Observation - Discussion	- Observes shaded regions - Discusses in groups - Describes regions - Shares ideas
20 mins	**EXPLAIN**	- Draws a Venn diagram with two intersecting sets - Labels each region: - **P only (P - Q):** elements in P but not in Q - **Q only (Q - P):** elements in Q but not in P - **P ∩ Q:** elements in both P and Q (intersection) - **P ∪ Q:** all elements in P or Q or both (union) - **Complement of P (P'):** elements not in P - **Complement of Q (Q'):** elements not in Q - Shows how to describe shaded regions using set notation - Examples: Shade P ∩ Q, shade P ∪ Q, shade P only, shade Q only, shade (P ∪ Q)'	- Lecture method - Use of diagrams - Demonstration - Note-taking	- Listens attentively - Takes notes - Copies diagrams - Asks questions
10 mins	**ELABORATE**	- Provides blank Venn diagrams with shaded regions - Asks learners to describe each shaded region using set notation - Provides descriptions and asks learners to shade the corresponding regions - Guides learners through the activities	- Guided practice - Application - Drawing	- Describes shaded regions - Shades given regions - Asks questions - Completes work
5 mins	**EVALUATE**	- Asks oral questions: - "What does P only mean?"	- Oral assessment - Written assignment	- Answers oral questions - Copies class work

		- "What does P ∩ Q represent?" - "How do you describe the union of two sets?" - "What is the complement of P?" - Gives class work		
5 mins	**EXTEND**	- Tells learners to draw a Venn diagram and shade the complement of H - M - Asks them to practice describing shaded regions at home - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. Describe the shaded part in the Venn diagram below:
   (Teacher draws diagram with shaded intersection)
2. Describe the shaded part where only set A is shaded.
3. Draw a Venn diagram and shade A ∪ B.
4. Draw a Venn diagram and shade (A ∩ B)'.
5. What does the unshaded region in a Venn diagram represent?

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 4

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Representing Information and Solving Word Problems on Venn Diagrams (Unknown in Intersection)

COMPETENCES:

- **Subject Competence:** The learner represents given information on a Venn diagram and solves for unknown values in the intersection.
- **Language Competence:** The learner interprets word problems and translates them into set notation.

TEACHING METHODS/TECHNIQUES:

- Problem-solving
- Discussion
- Guided practice
- Drawing
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Represent word problems on a Venn diagram
2. Form equations from the Venn diagram
3. Solve for unknown values in the intersection
4. Find the number of elements in different regions

TEACHING/LEARNING MATERIALS:

- Chart showing sample Venn diagram problems
- Blank Venn diagrams
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Tells a story: "In a club of 40 farmers, 28 grow maize, 13 grow rice, k farmers grow both, and 8 grow none." - Asks: "How can we organize this information?" - Introduces topic: Venn diagram word problems	- Storytelling - Question and answer - Brainstorming	- Responds to greeting - Listens to story - Suggests using Venn diagram - Listens
15 mins	**EXPLORE**	- Gives groups the problem: "In a class of 38 pupils, 22 like Math, 20 like English, r+2 like both, and 4 like neither." - Provides blank Venn diagram - Asks groups to place the given numbers in the correct regions - Asks: "What is unknown? How can we find it?" - Lets groups share their approaches	- Group work - Problem-solving - Discussion	- Works in groups - Places numbers on Venn diagram - Identifies unknown - Discusses solution method
20 mins	**EXPLAIN**	- Draws the Venn diagram on board with sets M and E - Labels regions: - M only: 22 - (r+2) - E only: 20 - (r+2) - Both: r+2 - Neither: 4 - Total: 38 - Forms equation: (22 - (r+2)) + (r+2) + (20 - (r+2)) + 4 = 38	- Lecture method - Step-by-step solution - Equation formation - Note-taking	- Listens attentively - Takes notes - Follows steps - Asks questions

		- Simplifies: 22 - r - 2 + r + 2 + 20 - r - 2 + 4 = 38 (22 - 2 + 2 + 20 - 2 + 4) + (-r + r - r) = 38 44 - r = 38 r = 6 - Therefore, both = r+2 = 8 pupils - Shows how to find M only, E only, etc.		
10 mins	**ELABORATE**	- Provides another problem: "In a club of 50 farmers, 27 grow millet, 25 grow cassava, n grow both, and 9 grow none." - Guides learners to set up the equation and solve for n - Asks them to find number of farmers who grow only one type of crop - Works through the solution step by step	- Guided practice - Application - Problem-solving	- Sets up equation - Solves for n - Finds only one crop - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "When the unknown is in the intersection, how do we set up the equation?" - "In the class problem, what was the value of r?" - "How many pupils liked only Math?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to practice more problems from the textbook - Asks them to create their own word problem and solve it - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. In a club of 50 farmers, 27 grow millet (M), 25 grow cassava (C), n grow both millet and cassava while 9 farmers do not grow any of the two. Find:
   a) The value of n
   b) The number of farmers who grow only one type of crop
2. In a class of 45 pupils, all of them eat beans, 25 eat fish, 15 eat chicken, k pupils eat all the three types of food while 10 eat beans only. Find the value of k.

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 5

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Venn Diagram Problems (When One of the Sets is Unknown)

COMPETENCES:

- **Subject Competence:** The learner solves Venn diagram problems where one of the sets is unknown.
- **Language Competence:** The learner forms and solves equations from Venn diagram information.

TEACHING METHODS/TECHNIQUES:

- Problem-solving
- Discussion
- Guided practice
- Equation formation
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Represent information where one set is unknown on a Venn diagram
2. Form equations to find the unknown
3. Calculate the number of elements in each region
4. Find probabilities from Venn diagram information

TEACHING/LEARNING MATERIALS:

- Chart with sample problems
- Blank Venn diagrams
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Presents problem: "In a team of 57 players, 22 like music, 2y like swimming, 11 like both, and 4 like neither." - Asks: "What is unknown here?" (2y, the number who like swimming) - Introduces topic: finding unknown sets	- Question and answer - Problem presentation	- Responds to greeting - Identifies the unknown - Listens
15 mins	**EXPLORE**	- Gives groups the problem to discuss - Provides blank Venn diagram - Asks them to place known values and express unknown regions in terms of y - Asks: "How can we form an equation to find y?" - Lets groups share their approaches	- Group work - Problem-solving - Discussion	- Works in groups - Places values on diagram - Expresses unknowns - Discusses equation

20 mins	**EXPLAIN**	- Draws Venn diagram with sets M and S - Labels regions: - M only: 22 - 11 = 11 - S only: 2y - 11 - Both: 11 - Neither: 4 - Total: 57 - Forms equation: 11 + (2y - 11) + 11 + 4 = 57 11 - 11 + 11 + 4 + 2y = 57 15 + 2y = 57 2y = 42 y = 21 - Therefore, S only = 2y - 11 = 42 - 11 = 31 players - Finds probability of picking a player who likes only one activity: Only one activity = M only + S only = 11 + 31 = 42 Probability = 42/57 = 14/19	- Lecture method - Step-by-step solution - Equation formation - Note-taking	- Listens attentively - Takes notes - Follows steps - Asks questions
10 mins	**ELABORATE**	- Provides another problem: "Of 50 farmers, 20 grow millet, p grow cassava, 8 grow both, and 6 grow none." - Guides learners to set up equation and find p and cassava only - Works through the solution step by step	- Guided practice - Application - Problem-solving	- Sets up equation - Solves for p - Finds cassava only - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "How do we handle problems where one set is unknown?" - "In the music/swimming problem, what was the value of y?" - "How many players liked only swimming?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to practice more problems with unknown sets - Asks them to complete Exercise 5 from the textbook - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

CLASS WORK:

1. In a team of 57 players, 22 like music, 2y like swimming, 11 like both, and 4 like neither. Find:
   a) The value of y
   b) The number of players who like swimming only
   c) The probability of picking a player who likes only one activity
2. Of 50 farmers, 20 grow millet, p grow cassava, 8 grow both, and 6 grow none. Find:
   a) The value of p
   b) The number of farmers who grow cassava only

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 6

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Venn Diagram Problems (When Given Complement or Difference of Sets as Unknown)

COMPETENCES:

- **Subject Competence:** The learner solves Venn diagram problems where the complement or difference is unknown.
- **Language Competence:** The learner interprets complement and difference in set notation.

TEACHING METHODS/TECHNIQUES:

- Problem-solving
- Discussion
- Guided practice
- Equation formation
- Note-taking

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. Represent problems with complement or difference on a Venn diagram
2. Form equations involving complements
3. Solve for unknown values
4. Calculate probabilities from the Venn diagram

TEACHING/LEARNING MATERIALS:

- Chart with sample problems
- Blank Venn diagrams
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Presents problem: "In a class of 60 pupils, 30 like English, y like Math only, 10 like both, and 5 like neither." - Asks: "What does 'Math only' mean? How is it different from 'like Math'?" - Introduces topic: problems with complements and differences	- Question and answer - Problem presentation	- Responds to greeting - Understands "only" - Listens
15 mins	**EXPLORE**	- Gives groups the problem to discuss - Provides blank Venn diagram - Asks them to place known values and express unknown regions - Asks: "How can we find y?" - Lets groups share their approaches	- Group work - Problem-solving - Discussion	- Works in groups - Places values on diagram - Expresses unknowns - Discusses equation
20 mins	**EXPLAIN**	- Draws Venn diagram with sets E and M - Labels regions: - E only: 30 - 10 = 20 - M only: y - Both: 10 - Neither: 5 - Total: 60 - Forms equation: 20 + y + 10 + 5 = 60 35 + y = 60 y = 25 - Therefore, M only = 25 pupils - Finds only one subject: 20 + 25 = 45 pupils - Presents second problem: "60 guests, 28 served Miranda, 24 served Fanta, 10 both, 7p+4 served none." - Forms equation: (28-10) + (24-10) + 10 + (7p+4) = 60 18 + 14 + 10 + 7p + 4 = 60 46 + 7p = 60 7p = 14	- Lecture method - Step-by-step solution - Equation formation - Note-taking	- Listens attentively - Takes notes - Follows steps - Asks questions

		p = 2 - Finds none = 7(2)+4 = 18 guests - Finds probability = 18/60 = 3/10		
10 mins	**ELABORATE**	- Provides another problem: "In a class of 57, 20 took Novida, 2m+2 took Pepsi only, 9 both, 3 took none." - Guides learners to set up equation and find m and Pepsi only - Works through the solution step by step	- Guided practice - Application - Problem-solving	- Sets up equation - Solves for m - Finds Pepsi only - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "How do we handle 'only' in Venn diagrams?" - "In the first problem, what was the value of y?" - "How many pupils liked only Math?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to practice more problems with complements - Asks them to complete Exercise 6 from the textbook - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. In a class of 60 pupils, 30 like English, y like Mathematics only, 10 like both subjects and 5 do not like any of the two subjects. Find:
   a) The value of y
   b) The number of pupils who like only one subject

2. At a party attended by 70 guests, 40 took beer, 30 took soda, 2p drinks both, and 12 guests did not take any of the two drinks. Find:
   a) The value of p
   b) The number of guests who took only one type of drink
   c) The probability that a guest took only soda

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 7

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Number Patterns	Solving Word Problems on Venn Diagrams by Comparing Regions

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COMPETENCES:

- **Subject Competence:** The learner solves Venn diagram problems by comparing given conditions.
- **Language Competence:** The learner interprets comparative statements and forms equations.

TEACHING METHODS/TECHNIQUES:

- Problem-solving
- Discussion
- Guided practice
- Equation formation
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Interpret comparative statements in word problems
2. Form equations by comparing regions
3. Solve for unknown values
4. Find probabilities from the Venn diagram

TEACHING/LEARNING MATERIALS:

- Chart with sample problems
- Blank Venn diagrams
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Presents Venn diagram with values: M region: 9, 2p, p; S region: p, p, 3 - Asks: "If 16 more learners passed Math than Science, how can we express this?" - Introduces topic: comparing regions	- Question and answer - Diagram presentation	- Responds to greeting - Observes diagram - Thinks about comparison - Listens
15 mins	**EXPLORE**	- Gives groups the problem with the Venn diagram - Asks them to write expressions for n(M) and n(S)	- Group work - Problem-solving - Discussion	- Works in groups - Writes expressions

		- Asks them to form an equation based on "16 more passed Math than Science" - Lets groups share their equations and solutions		- Forms equation - Solves for p
20 mins	**EXPLAIN**	- Draws the Venn diagram with sets M and S - Writes expressions: n(M) = 9 + 2p + p = 9 + 3p n(S) = p + p + 3 = 2p + 3 - Forms equation: n(M) - n(S) = 16 (9 + 3p) - (2p + 3) = 16 9 + 3p - 2p - 3 = 16 6 + p = 16 p = 10 - Finds n(M') = those who did not pass Math = (p+3) + (p+1) = 10+3 + 10+1 = 24 - Finds only one subject = (9+2p) + (p+3) = (9+20) + (10+3) = 29 + 13 = 42 - Finds probability = 42/63 = 2/3 - Presents second problem: "In a bus, 22 bought chicken only, d bought both, 2d-20 bought liver only, d-4 bought neither." If liver buyers are twice those who bought neither, find d.	- Lecture method - Step-by-step solution - Equation formation - Note-taking	- Listens attentively - Takes notes - Follows steps - Asks questions
10 mins	**ELABORATE**	- Guides learners to solve the second problem step by step - Forms equation: n(L) = d + (2d-20); none = d-4; condition: n(L) = 2 × none d + 2d - 20 = 2(d-4) 3d - 20 = 2d - 8 3d - 2d = 20 - 8 d = 12 - Finds total passengers: 22 + d + (2d-20) + (d-4) = 22+12+(24-20)+(12-4) = 22+12+4+8 = 46	- Guided practice - Application - Problem-solving	- Participates in solving - Forms equation - Finds d - Finds total
5 mins	**EVALUATE**	- Asks oral questions: - "How do we form equations when comparing regions?" - "In the first problem, what was the value of p?" - "How many learners did not pass Math?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work

5 mins	**EXTEND**	- Tells learners to practice more comparison problems - Asks them to complete Exercise 7 from the textbook - Prepares for next lesson on probability	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. In a certain class, 27 pupils play football only, (y+10) play basketball only, (y+5) play both, and 4 play other games. If 29 pupils play basketball, find:
   a) The value of y
   b) The number of pupils in the class
2. In a village, 34 farmers grow maize, k farmers grow rice only, 8 farmers grow both, and 3 grow neither. If 23 farmers do not grow maize at all, find k.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 8

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Seven	Mathematics		60 Minutes

TOPIC	**SUB-TOPIC**	**CONTENT**
Sets	Probability	Introduction to Probability

COMPETENCES:

- **Subject Competence:** The learner defines probability and calculates simple probabilities.

- **Language Competence:** The learner uses terms like probability, event, sample space, expected chances correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Discussion
- Guided practice
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Define probability
2. Identify sample space and expected chances
3. Calculate probability of simple events
4. Find probability of complementary events

TEACHING/LEARNING MATERIALS:

- Dice
- Coin
- Chart with probability examples
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Tosses a coin and asks: "What are the chances of getting a head?" - Rolls a dice and asks: "What are the chances of getting a 3?" - Asks: "How do we express chance mathematically?" - Introduces topic: probability	- Demonstration - Question and answer - Brainstorming	- Responds to greeting - Observes coin toss - Observes dice roll - Answers questions
15 mins	**EXPLORE**	- Gives groups dice and coins - Asks them to list all possible outcomes when tossing a coin (H, T) - Asks them to list all possible outcomes when rolling a dice (1,2,3,4,5,6) - Asks: "What is the chance of getting a head? A tail? An even number?" - Lets groups share their findings	- Group work - Hands-on activity - Discovery method	- Tosses coins - Rolls dice - Lists outcomes - Discusses chances
20 mins	**EXPLAIN**	- Defines **probability**: the number used to describe the chance that something will happen or not happen; the likelihood of an event to happen - Formula: Probability = number of expected chances / number of total chances P(E) = n(E.C) / n(T.C) - Example 1: If a dice is tossed once, find probability of a prime number T.C = {1,2,3,4,5,6} = 6 E.C = {2,3,5} = 3 Probability = 3/6 = 1/2 - Example 2: Probability of going on a day starting with "T" (Tuesday, Thursday) Total days = 7, Expected = 2, Probability = 2/7 - Example 3: Basket with 4 apples, 5 mangoes, probability of picking a mango Total = 9, Mangoes = 5, Probability = 5/9 - Explains **complementary events**: P(not happening) = 1 - P(happening) Example: If P(visit) = 1/2, then P(not visit) = 1 - 1/2 = 1/2 Example: In a football match, outcomes = {win, loss, draw}, P(win) = 1/3	- Lecture method - Formula explanation - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies formulas - Asks questions
10 mins	**ELABORATE**	- Provides practice problems: a) Probability of even number on dice b) Probability of number less than 5 on dice c) Probability of composite number on dice	- Guided practice - Application - Problem-solving	- Solves problems - Shows working

		d) Probability of going on a day starting with "S" e) Probability of wedding in a month with 30 days - Guides learners through solutions		- Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "What is probability?" - "What is the formula for probability?" - "If a dice is tossed, what is the probability of getting a 4?" - "If P(rain) = 3/4, what is P(no rain)?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to find the probability of picking a red card from a deck of cards - Asks them to practice more probability problems - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. If a dice is tossed once, what is the probability that the following will show on top?
   a) A prime number
   b) An even number
   c) A number less than 5
   d) A composite number
2. Juma is planning to travel to America next week. What is the probability that he will go on a day that starts with letter "S"?
3. Gerald's wedding is next year. What is the probability that it will be held in a month with 30 days?
4. In a basket, there are 4 oranges, 5 lemons and 7 apples. Find the probability of picking:
   a) An orange at random
   b) A lemon at random
5. The probability that Moses will sit for an examination is 3/4. What is the probability that he will not sit for the examination?

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

LESSON PLAN 9

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Sets	Probability	Probability from Venn Diagrams

COMPETENCES:

- **Subject Competence:** The learner calculates probabilities using information from Venn diagrams.
- **Language Competence:** The learner interprets Venn diagrams to find probabilities.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Discussion
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Read information from Venn diagrams
2. Calculate probabilities of events from Venn diagrams
3. Find probabilities of combined events

TEACHING/LEARNING MATERIALS:

- Chart showing Venn diagrams
- Blank Venn diagrams
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Draws a Venn diagram with numbers in each region - Asks: "How many people are in set F? In set C? In both?" - Asks: "If we pick one person at random, what is the chance they are in set F?" - Introduces topic: probability from Venn diagrams	- Question and answer - Diagram presentation	- Responds to greeting - Observes diagram - Answers questions - Listens
15 mins	**EXPLORE**	- Gives groups a Venn diagram with values: F only = 15, C only = 25, both = 10, neither = 5 - Asks them to find total number of people - Asks them to find probability of picking someone who eats fish, who eats chicken, who eats only one type - Lets groups share their answers	- Group work - Problem-solving - Discussion	- Works in groups - Finds total - Calculates probabilities - Shares answers

20 mins	**EXPLAIN**	- Draws Venn diagram with sets F (fish) and C (chicken) - Shows values: F only = 15, C only = 25, both = 10, neither = 5 - Finds total = 15 + 10 + 25 + 5 = 55 - Calculates probabilities: a) P(eats fish) = (15 + 10)/55 = 25/55 = 5/11 b) P(eats chicken) = (25 + 10)/55 = 35/55 = 7/11 c) P(eats only one type) = (15 + 25)/55 = 40/55 = 8/11 d) P(eats both) = 10/55 = 2/11 e) P(eats neither) = 5/55 = 1/11 - Shows how to express probabilities in simplest form	- Lecture method - Step-by-step calculation - Note-taking	- Listens attentively - Takes notes - Follows calculations - Asks questions
10 mins	**ELABORATE**	- Provides another Venn diagram problem: Farmers growing food crops (F) and cash crops (C): F only = 20, C only = 30, both = 15, neither = 10 - Asks learners to find: a) P(grows food crops) b) P(grows cash crops) c) P(grows only one crop) d) P(grows both) - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Finds total - Calculates probabilities - Participates - Asks questions
5 mins	**EVALUATE**	- Asks oral questions: - "How do we find total from a Venn diagram?" - "How do we find probability of an event from a Venn diagram?" - "In the fish/chicken example, what was P(eats fish)?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to create their own Venn diagram problem and find probabilities - Asks them to practice more problems - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

The Venn diagram below shows the number of farmers who grow food crops (F) and cash crops (C).

(Teacher draws diagram with: F only = 20, C only = 30, both = 15, neither = 10)

Find:

a) The total number of farmers

b) The probability of picking a farmer at random who grows food crops

c) The probability of picking a farmer at random who grows cash crops

d) The probability of picking a farmer at random who grows only one type of crop

e) The probability of picking a farmer at random who grows both crops

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LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 10

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Seven	Mathematics		60 Minutes

TOPIC	**SUB-TOPIC**	**CONTENT**
Numeration and Place Value	Whole Numbers	Place Value and Value of Whole Numbers

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COMPETENCES:

- **Subject Competence:** The learner identifies place value and value of digits in whole numbers.
- **Language Competence:** The learner uses terms like place value, value, digit correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Use of place value chart
- Note-taking

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. Identify the place value of digits in whole numbers
2. Calculate the value of digits in whole numbers
3. Use place value chart correctly
4. Solve problems involving place value and value

TEACHING/LEARNING MATERIALS:

- Place value chart (Th, H, T, O)
- Flash cards with numbers
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes number 249,367 on the board - Asks: "What does the digit 9 represent in this number?" - Asks: "What about the digit 2?" - Introduces topic: place value and value	- Question and answer - Brainstorming	- Responds to greeting - Observes number - Answers questions - Listens
15 mins	**EXPLORE**	- Gives groups different large numbers - Asks them to identify the place value of each digit using a place value chart - Asks them to find the value of specific digits - Lets groups share their findings	- Group work - Guided discovery - Use of chart	- Works in groups - Uses place value chart - Identifies place values - Finds values
20 mins	**EXPLAIN**	- Draws place value chart on board: Millions, Hundred Thousands, Ten Thousands, Thousands, Hundreds, Tens, Ones - Example 1: 249,367 - Place value of 9 is Thousands - Value of 9 = 9 × 1000 = 9,000 - Example 2: 82,439,507 - Place value of 2 is Millions - Value of 2 = 2 × 1,000,000 = 2,000,000 - Explains: Value = digit × place value - Example 3: If value of 6 is 6000, place value = 6000 ÷ 6 = 1000 (Thousands) - Example 4: If place value of 5 is millions, value = 5 × 1,000,000 = 5,000,000	- Lecture method - Use of chart - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions
10 mins	**ELABORATE**	- Provides practice problems: a) Place value of 7 in 5,678,432 b) Value of 4 in 3,456,789 c) If value of 3 is 300, find its place value d) If place value of 8 is ten thousands, find its value - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Solves problems - Shows working - Asks questions - Participates

5 mins	**EVALUATE**	- Asks oral questions: - "What is the place value of 9 in 249,367?" - "What is the value of 6 in 46,293?" - "If value of 5 is 5000, what is its place value?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to find the place value and value of digits in 947,856 - Asks them to practice more problems at home - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. What is the place value of 9 in 249,367?
2. Write the place value of 2 in 82,439,507.
3. The value of 6 in a number is 6000. What is its place value?
4. Find the value of 5 in a number if its place value is millions.
5. Given 947,856:
   a) Find the product of the place value of 4 and the value of 6.
   b) Work out the sum of the value of 9 and the value of 8.
   c) Calculate the difference between the place values of 7 and 5.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 11

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Numeration and Place Value	Whole Numbers	Writing Whole Numbers in Words and Figures

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COMPETENCES:

- **Subject Competence:** The learner writes whole numbers in words and figures correctly.
- **Language Competence:** The learner spells number words correctly and uses place value grouping.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Group work
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Write whole numbers in words
2. Write word names in figures
3. Form numbers from given digits
4. Find the largest and smallest numbers from given digits

TEACHING/LEARNING MATERIALS:

- Chart with number words
- Flash cards with numbers
- Place value chart
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes 92,407,013 on the board - Asks: "How do we read this number aloud?" - Asks learners to attempt reading it - Introduces topic: writing numbers in words	- Question and answer - Brainstorming	- Responds to greeting - Observes number - Attempts to read - Listens
15 mins	**EXPLORE**	- Gives groups different large numbers - Asks them to group the numbers into millions, thousands, units using commas - Asks them to write the numbers in words - Lets groups share their answers	- Group work - Guided discovery - Discussion	- Works in groups - Groups numbers - Writes in words - Shares answers
20 mins	**EXPLAIN**	013 Ninety-two million, four hundred seven thousand, thirteen - Example 2: 41,042 = Forty-one thousand, forty-two - Explains how to write word names in figures: - Forty-eight thousand thirty-six = 48,036	- Lecture method - Worked examples - Note-taking	Solves problems - Shows working - Asks questions - Participates

		- Seven hundred two thousand twenty-one = 702,021 - Ninety-one million one hundred thirty-four thousand nineteen = 91,134,019 - Shows how to form numbers from given digits: - Digits 3,4,2: all possible 3-digit numbers: 234,243,324,342,423,432 - Largest: 432, Smallest: 234 - Sum = 432 + 234 = 666		
10 mins	**ELABORATE**	- Provides practice problems: a) Write 76,000,012 in words b) Write in figures: "Ninety million, four hundred five thousand, six" c) Using digits 5,3,0, write all possible 3-digit numbers d) Using digits 4,3,1,9,8,7, write the largest number in words - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Solves problems - Shows working - Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "How do you write 55,009 in words?" - "How do you write 'seven hundred two thousand twenty-one' in figures?" - "What is the largest 3-digit number from digits 3,4,2?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to form the smallest four-digit number using digits 3,0,5,2 - Asks them to find the probability of writing an even number from digits 3,4,6 - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

CLASS WORK:

1. Write the following in words:
   a) 99,040
   b) 55,009
   c) 76,000,012

2. Write the following in figures:
   a) Forty-eight thousand thirty-six
   b) Seven hundred two thousand twenty-one
   c) Ninety-one million one hundred thirty-four thousand nineteen

3. Using the digits 3,4 and 2:
   a) Write down all the possible 3-digit numbers that can be formed
   b) Find the sum of the smallest and the largest numbers formed

4. Using the digits 5,3 and 0, write down all the 3-digit numbers formed.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 12

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Numeration and Place Value	Whole Numbers	Expanding Numbers Using Place Values

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COMPETENCES:

- **Subject Competence:** The learner expands numbers using place values.
- **Language Competence:** The learner uses terms like expanded form, place value correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Place value chart
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Expand numbers using place values
2. Write numbers in expanded form
3. Find the number from its expanded form

TEACHING/LEARNING MATERIALS:

- Place value chart
- Flash cards with expanded forms
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes 469,017 on the board - Asks: "What is the place value of each digit?" - Asks: "How can we write this number as a sum of its place values?" - Introduces topic: expanding numbers	- Question and answer - Brainstorming	- Responds to greeting - Identifies place values - Suggests expanded form - Listens
15 mins	**EXPLORE**	- Gives groups different numbers - Asks them to write each digit with its place value - Asks them to write the number in expanded form - Lets groups share their expanded forms	- Group work - Guided discovery - Discussion	- Works in groups - Identifies place values - Writes expanded form - Shares answers
20 mins	**EXPLAIN**	- Shows how to expand 469,017 using place values: Hundred Thousands = 400,000 + 60,000 + 9,000 + 0 + 10 + 7 = 400,000 + 60,000 + 9,000 + 10 + 7 - Example 2: Expand 571,620	- Lecture method - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions

		= 500,000 + 70,000 + 1,000 + 600 + 20 + 0 = 500,000 + 70,000 + 1,000 + 600 + 20 - Shows how to find the number from expanded form: (4×10,000) + (8×1,000) + (2×100) + (7×10) + (9×1) = 40,000 + 8,000 + 200 + 70 + 9 = 48,279 - Example with decimals: (6×100) + (3×10) + (9×1) + (4×0.01) = 600 + 30 + 9 + 0.0		
10 mins	**ELABORATE**	- Provides practice problems: a) Expand 345,681 using place values b) Expand 182.47 using place values c) Find the number: (8×1000)+(3×10)+(6×1)+(2×0.1) d) Find the number: (3×10)+(6×1)+(5×0.1)+(8×0.01) - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Solves problems - Shows working - Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "How do you expand 234,807?" - "What number is (5×100)+(9×1)+(4×0.1)+(7×0.01)?" - "What is 4 thousands + 8 hundreds + 2 ones in short?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to expand 74.831 using powers of ten - Asks them to practice more problems	- Home assignment	- Listens to home task - Prepares for next lesson

		at home - Prepares for next lesson					

CLASS WORK:

1. Expand the following numbers using place values:
   a) 345,681
   b) 583,241
   c) 789,240
   d) 182.47
2. Find the numbers which have been expanded to give:
   a) (8×1000) + (3×10) + (6×1) + (2×0.1)
   b) (5×100) + (9×1) + (4×0.1) + (7×0.01)
3. Write as a single number: (3×10) + (6×1) + (5×0.1) + (8×0.01)
4. Write in short: 4 thousands + 8 hundreds + 2 ones

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 13

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Numeration and Place Value	Whole Numbers	Expanding Numbers Using Powers of Ten

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COMPETENCES:

- **Subject Competence:** The learner expands numbers using powers of ten.
- **Language Competence:** The learner uses terms like power, exponent, base correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Place value chart
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Expand numbers using powers of ten
2. Write numbers in expanded form with powers of ten
3. Find the number from its expanded form with powers of ten

TEACHING/LEARNING MATERIALS:

- Chart with powers of ten
- Place value chart
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes 74.831 on the board - Asks: "How can we write this number using powers of ten?" - Reviews: $10^0 = 1$, $10^1 = 10$, $10^2 = 100$, $10^3 = 1000$, etc. - Introduces topic: expanding using powers of ten	- Question and answer - Review	- Responds to greeting - Reviews powers of ten - Listens
15 mins	**EXPLORE**	- Gives groups numbers to expand using powers of ten - Asks them to write each digit multiplied by the appropriate power of ten - Lets groups share their expanded forms	- Group work - Guided discovery - Discussion	- Works in groups - Writes expanded forms - Shares answers
20 mins	**EXPLAIN**	- Shows how to expand 74.831 using powers of ten: $7 \times 10^1 + 4 \times 10^0 + 8 \times 10^{-1} + 3 \times 10^{-2} + 1 \times 10^{-3}$ $= 70 + 4 + 0.8 + 0.03 + 0.001$ - Shows how to find number from expanded form: $(8\times10^4) + (3\times10^3) + (0\times10^2) + (6\times10^1) + (2\times10^0)$ $= 80,000 + 3,000 + 0 + 60 + 2 = 83,062$ - Example with decimals: $(3\times10^2) + (6\times10^0) + (7\times10^{-1}) + (2\times10^{-2})$ $= 300 + 6 + 0.7 + 0.02 = 306.72$	- Lecture method - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions
10 mins	**ELABORATE**	- Provides practice problems: a) Expand 234,865 using powers of ten b) Expand 24.0539 using powers of ten c) Find the number: $(4\times10^5)+(9\times10^4)+(1\times10^3)+(6\times10^2)+(3\times10^1)$	- Guided practice - Application - Problem-solving	- Solves problems - Shows working

		d) Find the number: $(5\times10^{2})+(1\times10^{0})+(3\times10^{-1})+(9\times10^{-2})$ - Guides learners through solutions		- Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "How do you expand 345.718 using powers of ten?" - "What number is $(2\times10^{1})+(8\times10^{0})+(4\times10^{-1})+(7\times10^{-2})$?" - "What is 10^{3}? 10^{-2}?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners to expand 129.36 using powers of ten - Asks them to practice more problems at home - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. Expand the following using powers of ten:
   a) 234,865
   b) 542,391
   c) 345.718
   d) 24.0539
   e) 129.36

2. Find the numbers which have been expanded below:
   a) $(4\times10^{5}) + (9\times10^{4}) + (1\times10^{3}) + (6\times10^{2}) + (3\times10^{1})$
   b) $(7\times10^{5}) + (2\times10^{4}) + (8\times10^{3}) + (6\times10^{2})$
   c) $(5\times10^{2}) + (1\times10^{0}) + (3\times10^{-1}) + (9\times10^{-2})$
   d) $(2\times10^{1}) + (8\times10^{0}) + (4\times10^{-1}) + (7\times10^{-2})$

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 14

DATE	CLASS	SUBJECT / L. AREA	NO. OF CHILDREN	TIME
	Primary Seven	Mathematics		60 Minutes

TOPIC	SUB-TOPIC	CONTENT
Numeration and Place Value	Whole Numbers	Rounding Off Whole Numbers

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COMPETENCES:

- **Subject Competence:** The learner rounds off whole numbers to the nearest tens, hundreds, thousands, etc.
- **Language Competence:** The learner uses terms like round off, approximate, estimate correctly.

TEACHING METHODS/TECHNIQUES:

- Demonstration
- Question and answer
- Guided practice
- Use of number line
- Note-taking

LEARNING OUTCOMES:

By the end of this lesson, learners should be able to:

1. Round off numbers to the nearest tens
2. Round off numbers to the nearest hundreds
3. Round off numbers to the nearest thousands

4. Round off numbers to the nearest ten thousands

TEACHING/LEARNING MATERIALS:

- Number line chart
- Flash cards with numbers
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes 4,936 on the board - Asks: "About how many is this? If we want to estimate, what could we round it to?" - Introduces topic: rounding off numbers	- Question and answer - Brainstorming	- Responds to greeting - Observes number - Suggests estimates - Listens
15 mins	**EXPLORE**	- Gives groups different numbers to round off to different place values - Asks them to decide whether to round up or round down based on the digit - Lets groups share their rounded numbers	- Group work - Guided discovery - Discussion	- Works in groups - Rounds off numbers - Discusses rules - Shares answers
20 mins	**EXPLAIN**	- Explains **rounding off**: a way of approximating a number to the nearest given place value - Shows rounding off scale: when digit is 5,6,7,8,9 → round up; when digit is 0,1,2,3,4 → round down - Example 1: Round off 493,612 to the nearest thousands - Underline thousands place (3), look at next digit (6) → 6 ≥ 5, so round up - 493,612 ≈ 494,000 - Example 2: Round off 6,419 to the nearest hundreds	- Lecture method - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions

		- Underline hundreds place (4), look at next digit (1) → 1 < 5, so round down - 6,419 ≈ 6,400 - Example 3: Round off 38,527 to nearest ten thousands - Underline ten thousands (3), look at next (8) → 8 ≥ 5, round up → 40,000 - Shows rounding to tens: 386 ≈ 390 (6 ≥ 5)		
10 mins	**ELABORATE**	- Provides practice problems: a) Round off 4594 to nearest tens b) Round off 3582 to nearest hundreds c) Round off 73,061 to nearest thousands d) Round off 21,469 to nearest ten thousands - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Solves problems - Shows working - Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "What is 386 rounded to the nearest tens?" - "What is 2,721 rounded to the nearest hundreds?" - "What is 52,941 rounded to the nearest thousands?" - "When do we round up?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners: A book costs sh. 27,500. If rounded to nearest thousands, will buyers be happy? Explain why. - Asks them to practice more rounding problems - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. Round off to the nearest tens:
   a) 386
   b) 4594
   c) 999

2. Round off to the nearest hundreds:
   a) 2721

b) 3582
c) 8457

3. Round off to the nearest thousands:
   a) 52,941
   b) 73,061
   c) 89,603

4. Round off to the nearest ten thousands:
   a) 95,032
   b) 21,469
   c) 16,347

5. A book costs sh. 27,500 at a shop. If the cost is rounded off to the nearest thousands, will the buyers be happy or not? Explain why.

LESSON EVALUATION:

STRENGTH:	**AREAS WHICH NEED IMPROVEMENT:**	**WAY FORWARD:**

LESSON PLAN 15

DATE	**CLASS**	**SUBJECT / L. AREA**	**NO. OF CHILDREN**	**TIME**
	Primary Seven	Mathematics		60 Minutes

TOPIC	**SUB-TOPIC**	**CONTENT**
Numeration and Place Value	Roman Numerals	Reading and Writing Roman Numerals

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COMPETENCES:

- **Subject Competence:** The learner reads and writes Roman numerals correctly.
- **Language Competence:** The learner uses Roman numeral symbols correctly.

TEACHING METHODS/TECHNIQUES:

- Use of charts
- Demonstration
- Question and answer
- Guided practice
- Note-taking

LEARNING OUTCOMES:
By the end of this lesson, learners should be able to:

1. Identify basic Roman numerals
2. Write numbers in Roman numerals
3. Convert Roman numerals to Hindu-Arabic numbers
4. Perform operations with Roman numerals

TEACHING/LEARNING MATERIALS:

- Chart showing Roman numerals
- Flash cards with Roman numerals
- Chalkboard/whiteboard
- Learners' books

LESSON DEVELOPMENT (5Es MODEL)

TIME	STAGE	TEACHER'S ACTIVITY	METHODS/TECHNIQUES	LEARNERS' ACTIVITY
5 mins	**ENGAGE**	- Greets learners - Writes I, V, X on the board - Asks: "What do these letters represent?" - Shows a clock face with Roman numerals - Introduces topic: Roman numerals	- Question and answer - Real-life connection	- Responds to greeting - Identifies symbols - Observes clock - Listens
15 mins	**EXPLORE**	- Gives groups chart with Roman numerals - Asks them to list the basic Roman numerals and their values - Asks them to try writing small numbers in Roman numerals - Lets groups share their findings	- Group work - Guided discovery - Discussion	- Works in groups - Lists basic numerals - Tries writing numbers - Shares answers
20 mins	**EXPLAIN**	- Presents table of **basic Roman numerals**: I = 1, V = 5, X = 10, L = 50, C = 100, D = 500, M = 1000 - Explains **repeated Roman numerals**: II=2, III=3, XX=20, XXX=30, CC=200, etc. (not more than three repeats) - Explains **subtractional Roman numerals**: IV = 4 (5-1), IX = 9 (10-1), XL = 40 (50-10), XC = 90 (100-10), CD = 400 (500-100), CM = 900 (1000-100) - Explains **additional Roman numerals**: VI = 6, VII = 7, VIII = 8, LX = 60, LXX = 70, LXXX = 80 - Example 1: Write 411 in Roman numerals 411 = 400 + 10 + 1 = CD + X + I = CDXI - Example 2: Write 1994 in Roman numerals 1994 = 1000 + 900 + 90 + 4 = M + CM + XC + IV = MCMXCIV - Example 3: Convert DCCCLXXXVIII to Hindu-Arabic DCCC = 800, LXXX = 80, VIII = 8 → 888	- Lecture method - Worked examples - Note-taking	- Listens attentively - Takes notes - Copies examples - Asks questions

10 mins	**ELABORATE**	- Provides practice problems: a) Write 54, 49, 99 in Roman numerals b) Convert LIX, XCIX, LXXXIV to Hindu-Arabic c) On a farm, there are 119 cows, 264 sheep, 186 goats. Find total in Roman numerals d) Alex sold 868 kg of maize. Write in Roman numerals - Guides learners through solutions	- Guided practice - Application - Problem-solving	- Solves problems - Shows working - Asks questions - Participates
5 mins	**EVALUATE**	- Asks oral questions: - "What is L in Roman numerals?" - "How do you write 49 in Roman numerals?" - "What is CDXLV in Hindu-Arabic?" - Gives class work	- Oral assessment - Written assignment	- Answers oral questions - Copies class work
5 mins	**EXTEND**	- Tells learners: Uganda got independence in MCMLXII. What year is that? - Asks them to practice more Roman numeral problems - Prepares for next lesson	- Home assignment	- Listens to home task - Prepares for next lesson

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CLASS WORK:

1. Write the following numbers in Roman numerals:
   a) 54
   b) 49
   c) 99
   d) 955
   e) 415
2. On a farm, there are 119 cows, 264 sheep and 186 goats. Express the total number of animals on the farm in Roman numerals.
3. Alex sold 868 kg of maize. Write the quantity of maize he sold in Roman numerals.
4. Write the following Roman numerals in Hindu-Arabic numerals:
   a) LIX
   b) XCIX

c) LXXXIV
d) MDCIV

5. Uganda got her independence in MCMLXII. Express this year in Hindu-Arabic numerals.

LESSON EVALUATION:

STRENGTH:	AREAS WHICH NEED IMPROVEMENT:	WAY FORWARD:

INCOMPLETE DOCUMENT CALL OR WHATSAPP 0750-470-900

TO PURCHASE FULL COPY AND MORE FROM P.1- P.7

SING: Mr. Tayson

Remaining topics include: Bases, Operations on Whole Numbers, Properties of Operations, Indices, Standard Form, Prime Factorization, GCF and LCM, Square and Cube Roots, Divisibility Tests, Number Sequences, etc.